

American Option Risk Surfaces Project

Detailed Student Handout, Claim Logic, Experiment Workflow, and Future Study Guide

SURF2026 Computational Finance Project

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1 How to Read This Handout

This document is a complete teaching and research handout for the American Option Risk Surfaces project. It is not a paper draft. Its purpose is to help students understand the full logic of the project: what was claimed, why each experiment was necessary, what the results mean, what claims were rejected, and how the work should be extended.

The project should be read as a sequence of increasingly disciplined research decisions:

1. Build and validate a numerical reference solver.
2. Use the solver to generate reliable American-option labels.
3. Test whether ordinary neural surrogates violate financial structure.
4. Replace weak claims with constrained, task-specific models.
5. Assemble a final workflow and test it against fresh finite-difference solves.
6. Translate the evidence into a cautious journal-facing claim.

The final research direction is:

A validated, modular, structure-preserving surrogate workflow for fast American-option risk surfaces.

The project is *not* a broad machine-learning algorithm comparison. It is also *not* a claim that a single neural network replaces a numerical American-option solver. The strongest contribution is the workflow discipline: each financial object is learned in a form consistent with its mathematical constraints.

2 Core Mathematical Problem

2.1 American Option Variational Inequality

Let $S > 0$ be spot, $K > 0$ strike, $\tau \in [0, T]$ time to maturity, r the risk-free rate, q the continuous dividend yield, and $\sigma > 0$ volatility. The payoff is

$$\Phi(S) = \begin{cases} (K - S)^+, & \text{put,} \\ (S - K)^+, & \text{call.} \end{cases}$$

Under the Black–Scholes model with continuous dividends, the American value $V(S, \tau)$ solves the variational inequality

$$V \geq \Phi, \quad \partial_\tau V - \mathcal{L}V \geq 0, \quad (V - \Phi)(\partial_\tau V - \mathcal{L}V) = 0,$$

where

$$\mathcal{L}V = \frac{1}{2}\sigma^2 S^2 V_{SS} + (r - q)SV_S - rV.$$

This system says three things:

- The option value can never be below payoff.
- In the continuation region, the Black–Scholes PDE holds.
- In the exercise region, value equals payoff and continuation stops.

The free boundary $B(\tau)$ separates exercise and continuation. For puts it is usually below strike. For dividend-paying calls, early exercise can occur above strike.

2.2 Continuation Premium

Define the continuation premium

$$C(S, \tau) = V(S, \tau) - \Phi(S).$$

For an American option, $C \geq 0$. This simple inequality becomes the key design rule for the price surrogate. Instead of predicting V directly, the final model predicts a nonnegative $\hat{C}$ and reconstructs

$$\hat{V} = \Phi + \hat{C}.$$

This guarantees $\hat{V} \geq \Phi$ by construction.

2.3 Why the Problem Is Hard for Surrogates

American options are harder than European options for surrogate modelling because:

- price is constrained by an obstacle;
- the exercise boundary is a separate object, not a standard smooth label;
- Greeks are least regular near the free boundary;
- dividend-paying calls create boundary regimes that differ from puts;
- random point splits can overstate performance because neighbouring points on the same PDE surface are strongly correlated.

The project therefore uses held-out regime tests, direct boundary diagnostics, and strict-mask Delta evaluation.

3 Final Claims and Their Logical Status

3.1 Claims Retained

Table 1: Final retained claims.

Claim	Status	Evidence and interpretation
Validated PSOR benchmark	Ready	The solver passes a European put closed-form check, a no-dividend American call theorem check, obstacle/complementarity diagnostics, and grid/domain checks. European put max error is 1.331×10^{-5} ; no-dividend call error is 5.318×10^{-11} .
Fast price and continuation-premium surfaces	Ready	The selected positive-premium model has held-out regime price RMSE 3.563×10^{-2} and zero price-obstacle violation. This supports a price/premium surface claim.

Claim	Status	Evidence and interpretation
Boundary diagnostics	Ready with scope	The direct option-specific call-augmented boundary head reduces call boundary MAE from 2.826×10^{-1} to 3.855×10^{-2} , a $7.33\times$ improvement. Blind dividend-call MAE is 3.050×10^{-2} . This supports diagnostic boundary estimates inside the tested regime envelope.
Delta diagnostics	Ready after reframing	Autograd Delta is weak, but a bounded supervised Delta head reduces held-out strict-mask Delta RMSE from 2.162×10^{-1} to 4.098×10^{-2} with zero bound violation. Delta should be claimed as a separate supervised risk head, not as a derivative of the price network.
Integrated application workflow	Ready	The final workflow combines price, boundary, and Delta heads. Against fresh PSOR solves, it achieves mean price RMSE 3.381×10^{-2} , mean boundary MAE 1.069×10^{-2} , mean strict Delta RMSE 2.671×10^{-2} , mean speedup 193.4, and zero price/Delta violations.

3.2 Claims Rejected or Restricted

Table 2: Claims that should not be made.

Weak claim	Why it fails	Correct wording
One neural network solves price, boundary, and Greeks	Price, boundary, and Delta have different numerical regularity. Autograd Delta from the price network is weak.	Use a modular workflow: price/premium surrogate, direct boundary head, bounded Delta head.
The price surrogate is a Greek engine	Strict-mask autograd Delta RMSE is 2.162×10^{-1} and remains 2.132×10^{-1} after derivative-aware fine-tuning.	Delta diagnostics require a separate bounded supervised head.
The boundary head is a universal free-boundary solver	Dividend-paying calls were weak until targeted augmentation was added.	Boundary estimates are credible diagnostics within the tested regime envelope.
The work proves broad ML algorithm superiority	The evidence compares selected surrogate designs, not all algorithms.	The contribution is a structure-preserving computational-finance workflow.

Weak claim	Why it fails	Correct wording
The surrogate replaces PSOR completely	The surrogate is calibrated from PSOR labels and must be spot-checked outside the training envelope.	Use the surrogate for fast scenario screening and PSOR for audit and new regimes.

4 Overall Workflow

4.1 High-Level Research Pipeline

The project pipeline is:

PSOR validation $\rightarrow$ risk analytics $\rightarrow$ dataset $\rightarrow$ surrogate price $\rightarrow$ boundary diagnostics
 $\rightarrow$ Delta diagnostics $\rightarrow$ integrated workflow $\rightarrow$ claim audit.

Each stage answers a different research question:

1. Can we trust the finite-difference solver?
2. Which regimes and financial quantities matter?
3. Can a surrogate preserve the American obstacle?
4. Can boundary information be learned reliably?
5. Can Greeks be obtained from the price network?
6. If not, what repair is empirically justified?
7. Does the final workflow work as an application engine?

4.2 Model Architecture Logic

Table 3: Final modular architecture.

Component	Input and output	Financial constraint
Price/premium surrogate	Input: $(\log(S/K), \tau/T, r, q, \sigma, T, \mathbf{1}_{\text{call}})$. Output: continuation premium $\hat{C} = \text{softplus}(f_{\theta}(x))$. Price is $\hat{V} = \Phi + \hat{C}$.	Guarantees $\hat{V} \geq \Phi$ and $\hat{C} \geq 0$.
Boundary head	Input: $(\tau/T, \sqrt{\tau/T}, r, q, r - q, \sigma, T, \mathbf{1}_{\text{call}})$. Output: $\hat{B}/K$.	Put boundary mapped to $(0, 1)$; dividend-call boundary mapped to $(1, \infty)$.
Delta head	Input: same as price surrogate. Output: $\hat{\Delta}$.	Call Delta mapped to $(0, 1)$; put Delta mapped to $(-1, 0)$.

5 Step-by-Step Experiment Guide

5.1 Experiment 01: Solver Validation

File: `experiments/01_solver_validation.py`

Purpose. Before training any surrogate, the project needs a reliable reference solver. Experiment 01 validates the Crank–Nicolson/PSOR implementation.

Method. It runs three checks:

1. European put closed-form comparison against Black–Scholes.
2. No-dividend American call comparison against the European call.
3. American put obstacle and boundary sanity check.

Key results.

- European put max error on finest grid: 1.331×10^{-5} .
- No-dividend American call max error: 5.318×10^{-11} .
- American put obstacle violation: 0.
- American put initial boundary: approximately $0.7444K$.

Conclusion. The solver is credible enough to generate labels. This experiment supports the “validated PSOR benchmark” claim.

5.2 Experiment 02: Grid and Domain Sensitivity

File: `experiments/02_convergence_domain.py`

Purpose. Check whether solver outputs are stable under grid refinement and domain extension.

Method. American puts and dividend-paying calls are solved on multiple grids and domains. The 480^2 grid acts as the reference for grid checks.

Key results.

- American put 240^2 vs. 480^2 price error: 5.746×10^{-5} .
- Dividend-call 240^2 vs. 480^2 price error: 4.440×10^{-5} .
- Domain checks produce stable initial boundaries.
- Obstacle violations remain zero.

Conclusion. The solver is not only correct in one validation case; it is stable across grid and domain settings.

5.3 Experiment 03: Stress Regime Maps

File: `experiments/03_stress_regime_maps.py`

Purpose. Understand how boundaries, Delta, Gamma, runtime, and PSOR iterations behave under financial stress.

Method. The experiment sweeps put volatility and call dividend yield, and constructs a call boundary heatmap over dividend and volatility.

Key results.

- Put boundary decreases as volatility rises: initial boundary moves from $0.811K$ at $\sigma = 0.2$ to $0.411K$ at $\sigma = 0.6$.
- Dividend-call boundary decreases as dividend yield rises: approximately $1.989K$ at $q = 0.03$ to $1.233K$ at $q = 0.14$ for the sweep.
- Higher volatility increases PSOR iterations and runtime.

Conclusion. The project should include stress regimes. It should not evaluate only easy baseline options.

5.4 Experiment 04: Quantitative Risk Analytics

File: `experiments/04_quantitative_risk_analytics.py`

Purpose. Turn qualitative stress maps into systematic risk analytics.

Method. The experiment sweeps volatility, rate, maturity, and dividend yield for puts and calls. It records boundary movement, Gamma concentration, PSOR iterations, and complementarity diagnostics.

Key results.

- Put boundary is highly sensitive to volatility and dividends.
- Dividend-call boundary is highly sensitive to dividend yield and volatility.
- Gamma concentration is strongest near the exercise boundary and varies strongly with stress parameters.

Conclusion. The surrogate should not only fit price; it should support boundary-aware risk-surface analytics.

5.5 Experiment 05: Surrogate Dataset Construction

File: `experiments/05_surrogate_dataset.py`

Purpose. Generate a reusable supervised-learning dataset from validated PSOR surfaces.

Dataset.

- 32 regimes.
- 140,800 spot-time samples.
- 70,400 puts and 70,400 calls.
- 24 training regimes and 8 held-out test regimes.
- Features: $(\log(S/K), \tau/T, r, q, \sigma, T, \mathbf{1}_{\text{call}})$.
- Labels: price, premium, payoff, boundary, exercise indicator, Greek mask, boundary mask, sample weights.

Important split logic. The regime test split withholds entire parameter regimes, not just random points. This prevents inflated performance from surface-neighbour correlations.

Conclusion. This dataset is the foundation for all surrogate experiments.

5.6 Experiment 06: Baseline Surrogates

File: `experiments/06_surrogate_baselines.py`

Purpose. Test ordinary neural surrogate designs and identify whether unconstrained models violate financial structure.

Models.

- Plain network.
- Obstacle-weighted network.
- Positive-premium network.
- Shape-regularized network.

Key results.

Model	Regime price RMSE	Max obstacle violation	Decision
Plain	8.115×10^{-2}	3.767×10^{-1}	reject
Obstacle-weighted	2.620×10^{-1}	4.199×10^{-1}	reject
Shape-regularized	2.466×10^{-1}	4.955×10^{-1}	reject
Positive premium	3.917×10^{-2}	0	retain

Conclusion. The positive-premium architecture is necessary. Penalizing obstacle violations in the loss is not enough; the output representation must enforce the constraint.

5.7 Experiment 07: Positive-Premium Tuning

File: `experiments/07_positive_premium_tuning.py`

Purpose. Tune the positive-premium architecture and investigate weighting near the boundary.

Key results.

- Best price RMSE model: `pp\_128\_boundary\_boost`, regime price RMSE about 3.617×10^{-2} .
- Best boundary-band RMSE model: `pp\_128\_weighted`, boundary RMSE about 2.078×10^{-2} .

Conclusion. Boundary-aware weighting improves performance in financially important regions. The final selection should consider both global price RMSE and near-boundary behaviour.

5.8 Experiment 08: Boundary-Distance Price Profiles

File: `experiments/08_boundary_distance_profiles.py`

Purpose. Select the final price/premium surrogate using boundary-distance diagnostics.

Key results.

- Selected model: `hard\_band`.
- Held-out regime price RMSE: 3.563×10^{-2} .
- Boundary-band RMSE: 1.792×10^{-2} .
- Max negative premium: 0.

- Max price obstacle violation: 0.

Conclusion. This is the final price model for the workflow.

5.9 Experiment 09: Delta Validation

File: experiments/09_delta_validation.py

Purpose. Test whether automatic differentiation through the price surrogate gives accurate Delta.

Method. Delta from autograd is compared with finite-difference Delta labels on the Greek mask and strict Greek mask.

Key results.

- Held-out regime strict Delta RMSE: 2.162×10^{-1} .
- Strict Delta MAE: 1.715×10^{-1} .
- 95% absolute Delta error: 4.170×10^{-1} .

Conclusion. Autograd Delta is too weak. The project must not claim that the price surrogate is a Greek engine.

5.10 Experiment 10: Application Risk-Surface Case Study

File: experiments/10_application_risk_surface_case_study.py

Purpose. Demonstrate application value of the price surrogate and test a simple premium-threshold boundary proxy.

Key results.

- Mean application price RMSE: about 3.384×10^{-2} .
- Mean speedup: about 309× in that experiment.
- Boundary threshold proxy is weak for exact boundary recovery.

Conclusion. The price/speed story is strong. The boundary proxy should be replaced by a direct boundary head.

5.11 Experiment 11: Direct Boundary Head Refinement

File: experiments/11_boundary_head_refinement.py

Purpose. Train direct boundary heads rather than inferring boundaries from premium thresholds.

Key results.

- Overall boundary MAE improved relative to threshold proxy.
- Put boundary MAE was acceptable, around 5.15×10^{-2} .
- Call boundary MAE remained weak, around 3.64×10^{-1} .

Conclusion. Direct boundary learning helps, but dividend-paying calls require targeted repair.

5.12 Experiment 12: Empirical Synthesis

File: `experiments/12_empirical_section_synthesis.py`

Purpose. Synthesize the first 11 experiments and decide which paper claims were ready.

Conclusion. At this stage, price/premium risk surfaces were ready. Boundary diagnostics were promising but weak for dividend calls. Delta was not ready. The paper should be framed as an application workflow, not an algorithm comparison.

5.13 Experiment 13: Option-Specific Dividend-Call Boundary Refinement

File: `experiments/13_option_specific_call_boundary_refinement.py`

Purpose. Repair the dividend-call boundary weakness with option-specific heads and targeted PSOR-generated dividend-call augmentation.

Key results.

- Original option-specific call MAE: 2.826×10^{-1} .
- Refined call-augmented call MAE: 3.855×10^{-2} .
- Improvement factor: $7.33\times$.
- Blind dividend-call test MAE: 3.050×10^{-2} .

Conclusion. The free-boundary diagnostic claim becomes credible after reframing and targeted augmentation.

5.14 Experiment 14: Target-Aligned Review

File: `experiments/14_target_aligned_review.py`

Purpose. Reassess the project after boundary repair and decide the journal-facing claim.

Conclusion. The paper should claim fast obstacle-consistent American option price and premium risk surfaces with scoped boundary diagnostics. It should not yet claim a Greek engine.

5.15 Experiment 15: Derivative-Aware Delta Refinement

File: `experiments/15_derivative_aware_delta_refinement.py`

Purpose. Test whether adding a Delta loss to the price surrogate repairs autograd Greeks.

Key results.

- Baseline strict Delta RMSE: 2.162×10^{-1} .
- Best derivative-aware strict Delta RMSE: 2.132×10^{-1} .
- Improvement factor: only $1.014\times$.
- Price RMSE improved slightly, but Delta remained weak.

Conclusion. Derivative-aware fine-tuning does not repair the Greek claim. The project must either demote Delta to future work or train a separate Delta head.

5.16 Experiment 16: Supervised Bounded Delta Head

File: experiments/16_supervised_delta_head.py

Purpose. Train a separate bounded Delta diagnostic head directly on finite-difference Delta labels.

Key results.

- Selected model: option_specific_delta_64.
- Held-out strict Delta RMSE: 4.098×10^{-2} .
- Strict Delta MAE: 3.110×10^{-2} .
- 95% absolute error: 8.736×10^{-2} .
- Improvement over autograd Delta: 5.28 $\times$.
- Max Delta bound violation: 0.

Conclusion. Delta can be claimed only after reframing: it is a separate supervised bounded diagnostic head, not the derivative of the price network.

5.17 Experiment 17: Claim Repair Synthesis

File: experiments/17_claim_repair_synthesis.py

Purpose. Summarize repaired claims and produce a manuscript-safe claim matrix.

Conclusion. The final paper claim should be modular:

PSOR benchmark + positive-premium price surrogate + boundary head + Delta head.

Algorithm-superiority and one-model-solves-all claims should be removed.

5.18 Experiment 18: Integrated Application Workflow

File: experiments/18_integrated_application_workflow.py

Purpose. Test the final assembled workflow against fresh PSOR solves.

Method. The workflow combines:

- price/premium surrogate from Experiment 08;
- option-specific boundary heads from Experiment 13;
- bounded Delta heads from Experiment 16.

It is evaluated on four fresh cases: baseline put, high-volatility dividend put, dividend-paying call, and high-dividend high-volatility call.

Key results.

Case	Price RMSE	Boundary MAE	Delta RMSE	Speedup
Put base	1.826×10^{-2}	7.610×10^{-3}	1.009×10^{-2}	85.1
Put stress	5.408×10^{-2}	9.202×10^{-3}	5.425×10^{-2}	415.9
Call base	1.860×10^{-2}	1.018×10^{-2}	8.411×10^{-3}	131.5

Call stress	4.430×10^{-2}	1.577×10^{-2}	3.409×10^{-2}	141.2
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Mean results.

- Mean price RMSE: 3.381×10^{-2} .
- Mean boundary MAE: 1.069×10^{-2} .
- Mean strict Delta RMSE: 2.671×10^{-2} .
- Mean speedup: 193.4×
- Max price obstacle violation: 0.
- Max Delta bound violation: 0.

Conclusion. This is the strongest application evidence. It demonstrates the final workflow as a complete fast risk-surface engine.

6 Logical Dependency Map

The project logic is not linear only in time; it is also a dependency graph.

Table 6: Logical dependencies between experiments.

Dependency	Required evidence	Why it matters
Solver validation → dataset	Experiments 01–02 must show solver correctness and stability.	If the solver is wrong, every surrogate label is wrong.
Stress analytics → dataset design	Experiments 03–04 reveal sensitivity to volatility, dividend, maturity, and boundary regimes.	The surrogate must include stress cases, not only easy baselines.
Dataset → price surrogate	Experiment 05 provides features, labels, splits, and weights.	Learning requires a controlled, reproducible dataset.
Baselines → positive-premium model	Experiment 06 shows unconstrained networks violate the obstacle.	The architecture must enforce financial structure.
Price model → Delta audit	Experiment 09 tests whether the price model derivative is useful.	A price model is not automatically a Greek model.
Boundary weakness → boundary repair	Experiments 10–12 show threshold and pooled boundary methods are weak for calls.	Dividend-call boundary learning needs direct option-specific augmentation.
Failed Delta repair → Delta head	Experiment 15 shows derivative-aware fine-tuning is insufficient.	A separate bounded Delta head is necessary.
Component repairs → integrated workflow	Experiments 13 and 16 repair boundary and Delta claims.	Only after these repairs can Experiment 18 test the full workflow.

7 Findings for Students

7.1 What Worked

1. **Validating the solver first worked.** It gave the project a credible numerical foundation.
2. **Continuation-premium prediction worked.** It converted the American obstacle from a penalty into a guaranteed structural property.
3. **Boundary-aware weighting helped.** The selected price model performed well near the free boundary.
4. **Direct boundary heads worked after augmentation.** The boundary task needed its own model and targeted dividend-call regimes.
5. **A supervised bounded Delta head worked.** This repaired the Delta claim after autograd Delta failed.
6. **Claim auditing worked.** The strongest final paper is narrower but much more credible.

7.2 What Failed

1. **Plain price regression failed.** It violated the payoff obstacle.
2. **Penalty-based obstacle handling failed.** Loss penalties did not guarantee financial admissibility.
3. **Premium-threshold boundary extraction failed as a boundary solver.** It was useful only as a rough diagnostic.
4. **Pooled boundary heads failed for dividend-paying calls.** Calls required option-specific modelling and targeted augmentation.
5. **Autograd Delta failed.** A smooth price surrogate did not provide accurate Greeks.
6. **Derivative-aware price fine-tuning failed to repair Delta.** Delta needed a separate head.

7.3 Most Important Research Lesson

The main lesson is:

For American options, different financial quantities require different surrogate structures.

Price, boundary, and Delta are mathematically linked, but they are not equally regular and should not be forced into one unconstrained network.

8 Recommended Methodology for Future Students

8.1 Before Running New Experiments

Students should always do the following:

1. Run the unit tests.

2. Re-run solver validation if changing numerical settings.
3. Check obstacle and complementarity diagnostics.
4. Define the exact claim being tested.
5. Decide in advance what metric would make the claim ready, weak, or rejected.

8.2 Standard Experiment Template

Every new experiment should answer:

1. What claim does this experiment test?
2. What data and model does it use?
3. What is the held-out test?
4. What metric decides success?
5. What failure mode is possible?
6. How will the paper wording change if the experiment succeeds?
7. How will the paper wording change if it fails?

8.3 Recommended Future Experiments

8.3.1 Future Experiment A: Larger Regime Envelope

Extend the dataset to more maturities, volatilities, dividends, and rates. The goal is not just better accuracy; it is to test whether the current workflow is stable outside the original 32 regimes.

Success criterion. Price obstacle and Delta bounds remain exact; price RMSE, boundary MAE, and Delta RMSE degrade smoothly rather than catastrophically.

8.3.2 Future Experiment B: Active Learning

Use surrogate disagreement or high error regions to decide where to run new PSOR solves. This is a natural next stage because PSOR labels are expensive relative to surrogate evaluation.

Workflow.

1. Train current workflow.
2. Generate candidate regimes.
3. Estimate uncertainty or disagreement.
4. Solve PSOR only where uncertainty is high.
5. Retrain and measure error reduction per new regime.

8.3.3 Future Experiment C: Shape Constraints

The price model currently enforces the obstacle, but not full monotonicity or convexity. Future work can add monotone or convex neural architectures or post-hoc diagnostics.

Warning. Penalty terms alone may fail, just as obstacle penalties failed. Prefer architectural constraints where possible.

8.3.4 Future Experiment D: Uncertainty Quantification

Train ensembles or conformal prediction intervals for price, boundary, and Delta. Use uncertainty to trigger PSOR spot checks.

8.3.5 Future Experiment E: Alternative Models

Extend from Black–Scholes to local volatility or stochastic volatility. This is much harder because the state dimension increases and the free boundary may not be represented by a simple curve.

8.4 Future Paper Direction

The most feasible journal paper remains:

A validated computational-finance workflow for structure-preserving American-option risk surfaces.

The paper should emphasize:

- validated numerical reference;
- obstacle-preserving price representation;
- direct boundary diagnostics;
- bounded Delta diagnostics;
- integrated speed/accuracy application evidence;
- honest limitations.

9 Reproducibility Guide

9.1 Core Commands

From the project root:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src \
python3 -m unittest discover -s tests
```

Compile all scripts:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src \
python3 -m compileall -q src tests experiments
```

Run an experiment:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src \
python3 experiments/18_integrated_application_workflow.py
```

Compile a report:

```
cd reports
latexmk -pdf -interaction=nonstopmode \
    experiment_18_integrated_application_workflow.tex
```


9.2 Important Output Files

- `results/experiment_05_surrogate_dataset/surrogate_dataset.npz`
Main supervised-learning dataset.
- `results/experiment_08_boundary_distance_profiles/selected_final_price_premium.pt`
Final price/premium surrogate.
- `results/experiment_13_option_specific_call_boundary/*.pt`
Final boundary heads.
- `results/experiment_16_supervised_delta_head/*.pt`
Final Delta heads.
- `results/experiment_18_integrated_application_workflow/integrated_case_metrics.csv`
Final integrated workflow metrics.
- `paper/sifin_manuscript.tex`
Current academic manuscript source.
- `output/pdf/student_handout_full_report.tex`
This student handout source.

10 Final Summary

The project began as a broad exploration of American option numerical surrogates. It became stronger after weak claims were removed. The final logic is:

1. A validated PSOR solver provides trustworthy labels.
2. A positive continuation-premium surrogate gives fast price and premium surfaces while enforcing the American obstacle.
3. Free-boundary estimation needs direct option-specific diagnostic heads, especially for dividend-paying calls.
4. Greeks should not be taken from price-network autograd without validation.
5. A bounded supervised Delta head repairs the Delta diagnostic claim.
6. The integrated workflow is fast and accurate enough for scenario screening, but it must remain paired with PSOR spot checks outside the calibrated regime envelope.

For students, the most important habit is to connect every claim to a specific experiment and every experiment to a possible failure mode. This project is valuable not because every model worked, but because the failures forced a more precise and credible research contribution.